Algorithm 2 Graphic Shapley Value Solver Algorithm

Input:: population $\mathcal{N}$, the number of Monte Carlo permutation sampling k, the size of negative population
Initialize $\phi = \mathbf{0}_{|\mathcal{N}|}$
for $(1, 2, \cdots, k)$ **do**
  $\pi \longleftarrow$ *Uniformly sample from* $\Pi_{\mathcal{C}}$, where $\Pi_{\mathcal{C}}$ is permutation set
  for $i \in \mathcal{N}$ **do**
    /* Obtain predecessors of player i in sampled permutation π */
    $S_\pi(i) \longleftarrow \{j \in \mathcal{N} | \pi(j) < \pi(i)\}$
    /* Update incompatibility weights */
    $\phi_i \longleftarrow \phi_i + \frac{1}{k}(v(S_\pi(i) \cup \{i\}) - v(S_\pi(i)))$
  end for
end for
$\phi \longleftarrow \phi / \sum \phi$
$\phi \longleftarrow (1 - \phi) / \sum (1 - \phi)$
Output:: ϕ

We take the inverse of WPG (Xing & Ghorbani, 2004) on the game graph as the metric σ. WPG is proposed to assess the popularity of a node in a complex network. The formula of WPG is given as follows:

$$\hat{\sigma}(u) = (1 - d) + d \sum_{v \in B(u)} \hat{\sigma}(v) \frac{I_u}{\sum_{p \in R(v)} I_p} \frac{O_u}{\sum_{p \in R(v)} O_p}, \tag{6}$$

where d is the damping factor set to 0.85, $B(u)$ is the set of nodes that point to u, $R(v)$ denotes the nodes to which v is linked, and I, O are the degrees of inward and outward of the node, respectively. Therefore, the metric σ evaluates how unpopular a node is and equals the inverse of the WPG value $\hat{\sigma}$.

Then we calculate the Shapley Value of each node by taking a characteristic function in equation 2, named the graphic Shapley Value. We use Monte Carlo permutation sampling (Castro, Gómez, & Tejada, 2009) to approximate the Shapley Value, which can reduce the computational complexity from exponential to linear time. After inverting the probabilities of the graphic Shapley Value, we get the cooperative incompatible distribution ϕ, where strategies that fail to collaborate with others have higher probabilities. The details are given in Algorithm 2.

Reward Solver for COLE$_\mathbf{R}$. COLE$_\mathrm{R}$ substitutes the Shapley value for WPG weighted average rewards within the population. Specifically, for each player $i \in \mathcal{N}$, we have

$$\phi_i = \sum_{j \in \mathcal{N}} \sigma(i)\sigma(j)\mathbf{w}(i, j), \tag{7}$$

where $\mathbf{w}(i, j)$ is the weight in the game graph, i.e., average payoffs of player i and j. σ is calculated as same as the calculation of graphic Shapley value solver for COLE$_\mathrm{SV}$. The remainder of the reward solver processes are identical to the graphic Shapley value solver.

5.2 Trainer: Approximating Best-Preferred Strategy

The trainer inputs the cooperative incompatible distribution ϕ and samples its teammates to learn to approach the best-preferred strategy against the IPI mixture.

Recall the oracle for $s_n : s_{n+1} = \text{oracle}(s_{n+1}, \mathcal{J}(s_n, \phi))$, with $\mathcal{R}(\eta(s_{n+1})) > k$. COLE_{SV} aims to optimize the best-preferred strategy over the IPI mixture. The $\mathcal{J}(s_n, \phi)$ is the joint objective that consists of individual and cooperative compatible objectives. The individual objective aims to improve performance within itself and promote adaptive ability with expert partners, formulated as follows: $\mathcal{J}_i(s_n) = \mathbf{w}(s_n, s_n)$, where s_n is the ego strategy that needs to be optimized in generation n.

And the cooperative compatible objective aims to improve cooperative outcomes with those failed-to-collaborate strategies: $\mathcal{J}_c = \mathbb{E}_{p\sim\phi}\mathbf{w}(s_n, p)$, where the objective is the expected rewards of s_n with cooperative incompatible distribution-supported partners. $\mathbf{w}$ estimates and records the mean cumulative rewards of multiple trajectories and starting positions. The expectation can be approximated as follows: $\mathcal{J}_c = \sum_1^b \phi(p^i)\mathbf{w}(s_t, p^i)$, where b is the number of sampling times.

To balance exploitation and exploration as the learning continues, we present the Sampled Upper Confidence Bound for Game Graph (SUCG) which combines the Upper Confidence Bound (UCB) and GFG to control the sampling for more strategies with higher probabilities or new strategies. Additionally, we view the SUCG value as the probability of sampling teammates instead of using the maximum item in typical UCB algorithms. Specifically, in the game graph, we keep the information on the times that a node has been visited. Therefore, the probability of each node considers both the Shapley Value and visiting times, denoted as $\hat{p}$. The SUCG for any node u in $\mathcal{N}$ could be calculated as follows:

$$\hat{\phi}(u) = \phi(u) + c\frac{\sqrt{\sum_{i\in\mathcal{N}} \mathbf{N}(i)}}{1 + \mathbf{N}(u)}, \tag{8}$$

where c is a hyperparameter that controls the degree of exploration and $\mathbf{N}(i)$ is the visit times of node i. SUCG could efficiently prevent COLE_{SV} from generating data with a few fixed strategies that did not cooperate, which could lead to loss of adaptive ability.

We conclude COLE_{SV} as Algorithm 1. Furthermore, to verify the influence of different ratios of two objectives, we denote COLE_{SV} with different ratios as 0:4, 1:3, 2:2, and 3:1. Specifically, COLE_{SV} with $a : b$ represents different partner sampling ratios for the combining objective, where a is the corresponding times to generate data using self-play for the individual objective, and b is the number of sampling times in $\mathcal{J}_c$. For example, COLE_{SV} 1:3 trains using self-play once and sampling from the cooperative incompatible distribution as partners three times to generate data and update the objectives.

6. Human-AI Experiment Pipeline

In this section, we present our Overcooked human-AI experiment pipeline, explicitly crafted for a comprehensive and streamlined assessment of the cooperative performance of AI agents, working in collaboration with novice human players. Fig. 4 illustrates the proposed experimental framework integrating human and AI participants, encompassing six crucial sequential stages. The crux of this pipeline lies in its human-AI evaluation design, integrating a broad

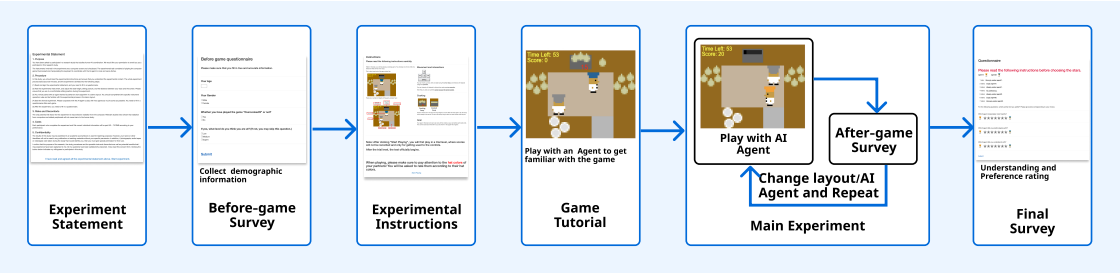

Figure 4: The illustrated figure characterizes the conceptual architecture of the proposed human-AI experimental pipeline, structured around six critical stages. **(Stage 1) Experiment Statement** delineates the nature of the experiment, associated risks, and ethical considerations among other relevant aspects. **(Stage 2) Before-game Survey** principally focuses on the acquisition of participant information. Delving deeper, **(Stage 3) Experimental Instructions** dispenses extensive procedural guidelines for the experiment. Subsequently, participants are invited to engage in a sequence of trial games to acquaint themselves with the experimental procedure in **(Stage 4) Game Tutorial**. Proceeding to **(Stage 5) Main Experiment**, it entails various rounds with a diverse array of differentiated AI agents. Post each round, participants are obliged to fill out a survey. The entire experimental process culminates with a comprehensive evaluation of the collaborative AI counterparts in **(Stage 6) Final Survey**. The pipeline is integrated into one platform designed for seamless interaction. Researchers have the flexibility to tailor the pipeline according to their needs, while participants benefit from a user-friendly interface that enables them to complete the stages with ease.

array of subjective metrics. These metrics scrutinize individual games involving AI agents, along with providing a comprehensive comparative analysis across the entire participant pool. These encompassing metrics primarily evaluate vital aspects such as intentionality, contribution, and teamwork within human-AI collaborations. Moreover, these metrics facilitate an extension of this evaluation to include parameters such as fluency, preference, and comprehension across all collaborating agents. Besides, all six steps are integrated into one platform, named COLE-Platform, designed for seamless interaction. Researchers have the flexibility to tailor the pipeline according to their needs, while participants benefit from a user-friendly interface that enables them to complete the stages with ease.

As shown in Fig. 4, the initial stage entails the formulation of an experiment statement, where the ethical considerations and potential risks associated with the experiment are thoroughly assessed, alongside the information of pertinent experimental details. Subsequently, participants are required to complete a preliminary survey, which seeks to gather fundamental information like their familiarity with the game and their age for more nuanced analyses. The next stage comprises detailed experimental instructions and a game tutorial, equipping participants with the necessary knowledge and skills to play the game and allowing them some initial exposure to the game for familiarization purposes. Then, the process